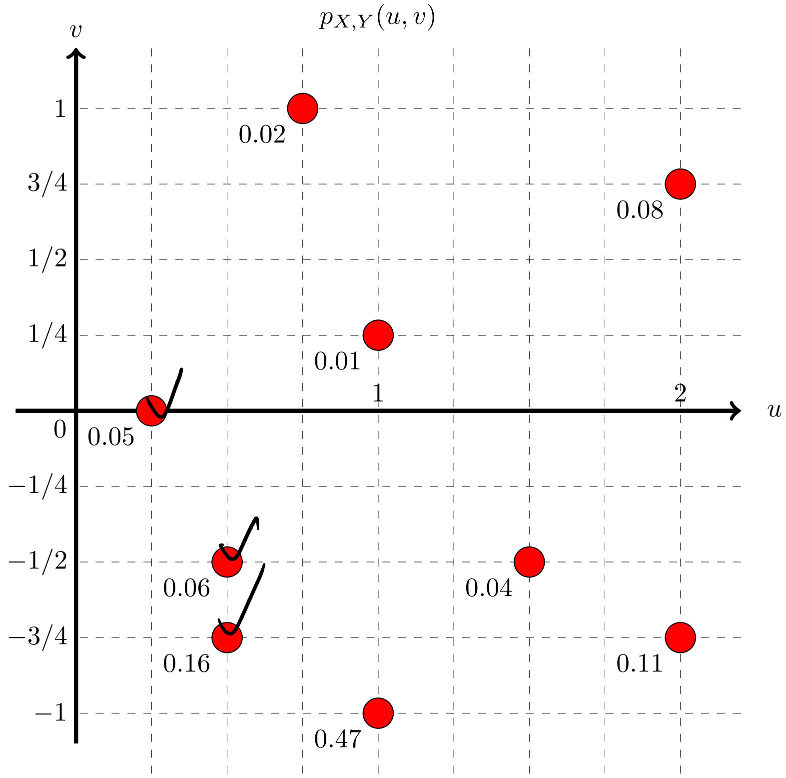
Random variables X and Y have the joint probability mass function shown below. What is the probability that $X^2 + Y^2$ is less than one?



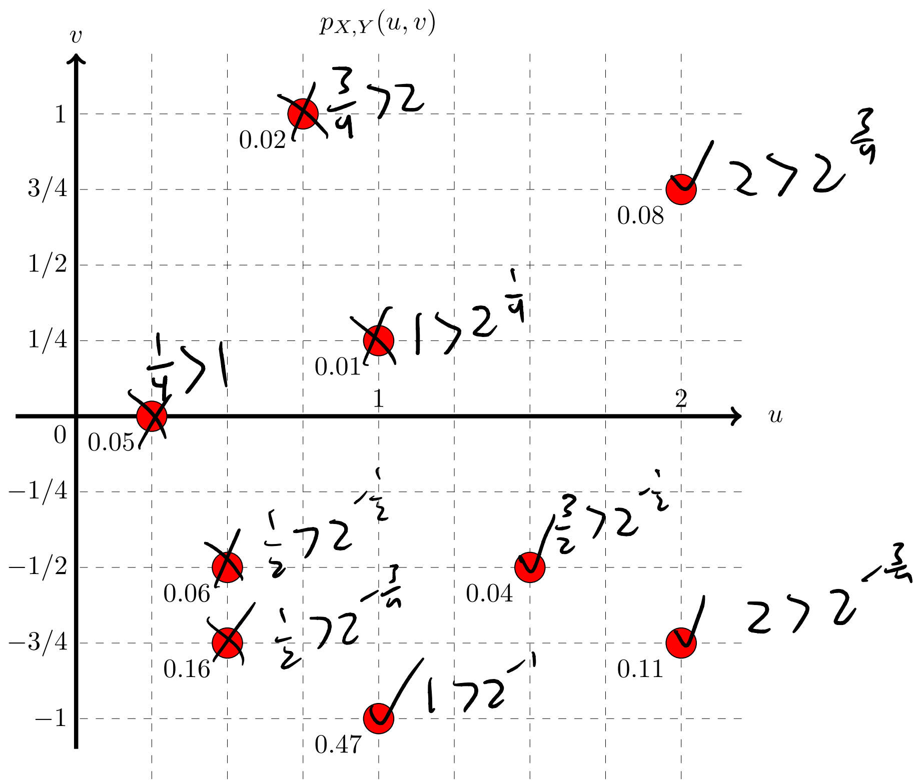
$P(X^2+Y^2<1)$

so $X<1$, $Y<1$

$= 0.27$

(a) 0.27
(b) 0.73
(c) 0.59
(d) 0.62
(e) 0.51
(f) 0.71
(g) 0.86
(h) 0.05
(i) 0.11
(j) 0.35
(k) 1
(l) 0
(m) None of these

Random variables X and Y have the joint probability mass function shown below. What is the probability that X is greater than 2^Y ?



$P(X > 2^Y)$

$.08 + .11 + .04 + .47$

(a) 0.7

(b) 0.59

(c) 0.62

(d) 0.51

(e) 0.71

(f) 0.86

(g) 0.92

(h) 0.75

(i) 0.35

(j) 1

(k) 0

(l) None of these

The joint probability density function of random variables X and Y is $f_{X,Y}(u,v) = 2e^{-u-2v}$ whenever $u, v > 0$, and is zero otherwise. What is the probability that $X+Y$ is less than 1 ?

(a) $(1-e^{-1})^2$

(b) $1-e^{-1}$

(c) $(1-e^{-2})^2$

(d) $1-e^{-2}$

(e) $1-e^{-1}-e^{-2}$

(f) $1/e$

(g) $1/2$

(h) $1/e^2$

(i) e^{-3}

(j) $1-e^{-3}$

(k) $1+2e^{-1}+e^{-2}$

(l) None of these

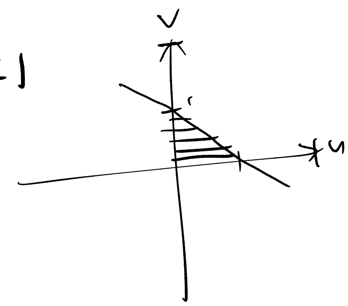
$P((X+Y)<1)$

$v<1-u$

$u<1-v$

$f_{x,y}(u,v) = 2e^{-u-2v}$ for $u,v>0$

$u+v<1$



$$\int_0^1\int_0^{1-v} 2e^{-u-2v}\,du\,dv$$

$$2\int_0^1\int_0^{1-v} e^{-u}e^{-2v}\,du\,dv$$

$$= 2\int_0^1 e^{-2v}\left[-e^{-u}\right]_0^{1-v} dv$$

$$= 2\int_0^1 e^{-2v}\left(-e^{-(1-v)}+1\right) dv$$

$$= 2\int_0^1 e^{-2v}\left(-\frac{1}{e}e^{v}+1\right) dv$$

$$= 2\int_0^1\left(-\frac{1}{e}e^{-v}+e^{-2v}\right) dv$$

$$= -\frac{2}{e}\int_0^1 e^{-v}\,dv + 2\int_0^1 e^{-2v}\,dv$$

$$= -\frac{2}{e}\left[-e^{-v}\right]_0^1 + \cancel{2}\left[-\cancel{\frac{1}{2}}e^{-2v}\right]_0^1$$

$$= -\frac{2}{e}\left(-\frac{1}{e}+1\right) - \left(\frac{1}{e^2}-1\right)$$

$$= \frac{2}{e^2} - \frac{2}{e} - \frac{1}{e^2} + 1$$

$$= \frac{1}{e^2} - \frac{2}{e} + 1$$

$$\left(1-\frac{1}{e}\right)^2 = \left(1-\frac{1}{e}\right)\left(1-\frac{1}{e}\right)$$

$$= 1 - \frac{1}{e} - \frac{1}{e} + \frac{1}{e^2}$$